AI CYBERSECURITY PROFESSIONAL

Independent Professional Training Program

60 hours (3 months) • AI Foundations • Secure AI • AI-Assisted Security • Automation • Governance • Hands-On Labs

Course Level	Beginner to Intermediate
Lab Philosophy	Free, open-source, local-first tools wherever practical
Learning Style	Theory + guided labs + security challenges + capstone
Primary Theme	Use AI safely to improve cybersecurity while learning how to protect AI-enabled systems

Course Positioning & Independence

This is an independent training program. It is not an official CompTIA course, does not reproduce CompTIA courseware, and is not affiliated with or endorsed by CompTIA.

The curriculum was informed by the publicly available CompTIA SecAI+ CY0-001 pre-draft objectives, which CompTIA itself describes as directional material subject to change. The course below deliberately uses original wording, a different sequence, additional hands-on activities and a free/local tool strategy.

Course Overview

AI is becoming part of security operations, development, threat analysis and enterprise workflows. This program teaches learners both sides of the problem: how to use AI as a cybersecurity assistant and how to secure AI applications, data, models, prompts, agents and integrations. Labs are designed around free/open-source software and local environments so learners can practice without requiring paid AI APIs.

Learning Outcomes

- Explain practical AI, machine-learning, LLM and generative-AI concepts from a cybersecurity perspective.
- Run a local LLM and understand model, prompt, context and inference concepts.
- Create safe prompts and reusable security-analysis prompt templates.
- Understand embeddings, vector search and retrieval-augmented generation (RAG).
- Build a small local RAG assistant over security documentation and synthetic data.
- Identify major AI application threats such as prompt injection, data leakage, insecure output handling and excessive agent authority.
- Design practical controls including input validation, output validation, access control, rate limits, logging and human approval.
- Use AI to assist log analysis, threat intelligence, vulnerability review, code review and incident documentation.
- Build Python-based security automation and simple AI-assisted workflows.
- Understand AI agents, tool calling, MCP-style integrations and least-privilege design.
- Perform basic AI security testing against intentionally vulnerable local applications.
- Apply responsible-AI, privacy, governance, risk and compliance concepts.
- Design monitoring and evaluation for AI security quality, accuracy and misuse.
- Complete an end-to-end AI cybersecurity capstone using only free/local tools.

Free / Open-Source Lab Stack

Area	Free / Local Tool
Local AI	Ollama + freely available local models such as Qwen, Llama-family or other compatible models
AI UI	Open WebUI or another free/local interface where appropriate
Programming	Python 3, VS Code / VSCodium, Jupyter
RAG	ChromaDB or FAISS, sentence-transformers
Security Testing	OWASP ZAP, Burp Suite Community Edition, Nmap
SOC / Monitoring	Wazuh, Wireshark, Sigma rule concepts
Threat Intelligence	MITRE ATT&CK, MISP Community Edition
AI Security Testing	Garak and local test harnesses where supported
Model/ML Utilities	Hugging Face libraries, scikit-learn
Data	CSV, JSON, synthetic security logs, public non-

	sensitive datasets
Automation	Python, Bash, PowerShell, n8n Community Edition where locally deployable
Containers	Docker Community Edition, Trivy
Code Security	Semgrep Community Edition, Bandit, Gitleaks
Documentation	LibreOffice

Tool availability and model hardware requirements vary. The instructor should select a model that runs on the learner's computer. Cloud APIs are optional and are not required for the core labs.

Recommended Lab Environment

- Windows or Linux host with Python 3 and VS Code/VSCodium.
- At least 16 GB RAM recommended for a comfortable local-LLM learning environment; smaller models may run on lower-spec systems.
- Ollama installed locally with one suitable open model.
- Optional Kali Linux VM for security testing.
- Optional Ubuntu VM for Wazuh, Docker and server-side labs.
- Isolated vulnerable web application such as OWASP Juice Shop for AI-assisted security testing.
- Synthetic logs and test credentials only.

Module	Title	Focus
01	AI & Cybersecurity Foundations	AI vocabulary, ML, GenAI, LLMs and security use cases
02	How Modern AI Systems Work	Training, inference, transformers, tokens, context and model limits
03	Local AI Lab with Ollama	Run models locally and establish a private AI workspace
04	Security Prompt Engineering	Prompt patterns, structured outputs and analyst workflows
05	AI Data Security	Data quality, provenance, privacy, leakage, masking and minimization
06	RAG for Cybersecurity	Embeddings, vector stores, retrieval and grounded answers
07	Secure AI Application Architecture	Model, application, gateway, data and user trust boundaries
08	AI Threat Modeling	Threat modeling AI apps using OWASP, MITRE ATLAS and risk thinking
09	Prompt Injection & AI Application Attacks	Injection, jailbreaks, output risks and indirect attacks
10	AI Access Control & Guardrails	Least privilege, permissions, quotas, validation and human approval
11	AI Monitoring, Logging & Evaluation	Quality, hallucination checks, audit trails and abuse monitoring
12	AI-Assisted SOC Operations	Alert triage, log analysis, threat hunting and investigation
13	AI-Assisted Vulnerability & Code Security	Secure coding, SAST, dependency review and vulnerability analysis
14	AI-Assisted Threat Intelligence	IOC extraction, enrichment, ATT&CK mapping and reporting
15	AI for Malware & Forensics	Safe triage, static evidence

		analysis and forensic summarization
16	Security Automation with Python	Build repeatable analyst automations
17	AI Agents & Tool Calling	Agent loops, tools, permissions, state and safe execution
18	MCP-Style Security Integrations	Connect AI assistants to safe read-only security tools
19	AI-Enhanced Attack Simulation	How AI can accelerate reconnaissance and social-engineering risk; defensive testing
20	AI Governance, Risk & Responsible Use	Policies, privacy, bias, transparency, accountability and risk
21	AI Security Operations Challenge	Investigate and defend an AI-enabled incident
22	Final Capstone: Secure AI Cybersecurity Assistant	Build, test, monitor and document a local AI security solution

Detailed Module Content

Module 01: AI & Cybersecurity Foundations

Theory Concepts

- Artificial intelligence vs machine learning vs generative AI
- Classification, clustering and anomaly-detection concepts
- LLMs, small language models and multimodal AI
- Where AI helps security teams and where human judgment remains essential
- AI attack surface vs traditional application attack surface

Hands-On Lab Objective

Create an AI-in-security use-case map for a small SOC.

Practical Lab Activities

1. Classify security use cases as detection, investigation, automation or reporting.
2. Identify where hallucination could create risk.
3. Create a human-approval boundary for each use case.
4. Write an AI risk/benefit table.

Free / Local Tools

Python, LibreOffice, Ollama

Expected Learning Outcome

Students can explain AI concepts in security language and identify suitable vs risky use cases.

Module 02: How Modern AI Systems Work

Theory Concepts

- Training vs inference
- Tokens, context windows and embeddings
- Transformer and attention concepts
- Supervised, unsupervised and reinforcement learning
- Fine-tuning, quantization and model selection
- Model evaluation and limitations

Hands-On Lab Objective

Run a local model and observe how prompt, context and model parameters affect output.

Practical Lab Activities

5. Install a local model.
6. Test deterministic and creative prompts.
7. Compare two small models if hardware permits.
8. Measure response quality against a fixed test set.
9. Document limitations.

Free / Local Tools

Ollama, Python, Jupyter

Expected Learning Outcome

Students understand the practical lifecycle from model selection to inference.

Module 03: Local AI Lab with Ollama

Theory Concepts

- Local inference architecture
- Model storage and access
- Local API concepts
- Resource management
- Basic model security

Hands-On Lab Objective

Create a private local AI workstation that can be used throughout the course.

Practical Lab Activities

10. Install Ollama.
11. Pull a suitable model.
12. Use the command line.
13. Call the local API from Python.
14. Create a basic security-assistant script.
15. Restrict the assistant to synthetic lab data.

Free / Local Tools

Ollama, Python, VS Code

Expected Learning Outcome

Students can run and program against a local AI model without paid API keys.

Module 04: Security Prompt Engineering

Theory Concepts

- System instructions and user instructions
- Zero-shot, one-shot and few-shot approaches
- Prompt templates
- Structured output and JSON
- Context injection risks
- Verification and human review

Hands-On Lab Objective

Build reusable prompts for a SOC analyst.

Practical Lab Activities

16. Create an alert-triage prompt.
17. Create an IOC-extraction prompt.
18. Force structured JSON output.
19. Test ambiguous input.
20. Compare AI output with known answers.
21. Add a verification checklist.

Free / Local Tools

Ollama, Python, JSON

Expected Learning Outcome

Students can create repeatable security prompts and validate AI output.

Module 05: AI Data Security

Theory Concepts

- Structured, semi-structured and unstructured data
- Data cleansing and validation
- Data provenance and lineage
- Sensitive information exposure
- Masking, redaction and minimization
- Integrity and synthetic-data generation

Hands-On Lab Objective

Build a safe preprocessing pipeline before sending security data to an AI model.

Practical Lab Activities

22. Create synthetic SOC logs.
23. Detect email/IP/user fields.
24. Mask sensitive values.
25. Validate data format.
26. Record source and transformation steps.
27. Compare original and sanitized data.

Free / Local Tools

Python, Pandas, Regex, LibreOffice

Expected Learning Outcome

Students can prepare security data for AI while reducing unnecessary disclosure.

Module 06: RAG for Cybersecurity

Theory Concepts

- Why retrieval improves grounding
- Embeddings and semantic similarity
- Vector databases
- Chunking and metadata
- Retrieval quality
- RAG hallucination risks

Hands-On Lab Objective

Build a local RAG assistant over cybersecurity notes and public/non-sensitive documents.

Practical Lab Activities

28. Create a small knowledge base.
29. Generate embeddings.
30. Store vectors in ChromaDB or FAISS.
31. Retrieve relevant passages.
32. Ask the local model to answer from retrieved context.
33. Add source references and an unknown-answer rule.

Free / Local Tools

Ollama, ChromaDB or FAISS, sentence-transformers, Python

Expected Learning Outcome

Students can build a basic grounded cybersecurity assistant entirely locally.

Module 07: Secure AI Application Architecture

Theory Concepts

- AI application components
- Trust boundaries
- Model gateway
- Data layer
- User and agent permissions
- Secrets and API security
- Failure handling

Hands-On Lab Objective

Draw and review a secure architecture for a local AI security assistant.

Practical Lab Activities

34. Create a data-flow diagram.
35. Identify trust boundaries.
36. Mark sensitive data paths.
37. Add authentication and authorization points.
38. Define safe failure behavior.
39. Document security controls.

Free / Local Tools

Draw.io or diagrams.net, LibreOffice

Expected Learning Outcome

Students can reason about the security architecture of AI-enabled applications.

Module 08: AI Threat Modeling

Theory Concepts

- Threat actors and assets
- AI-specific attack surfaces
- Abuse cases
- Risk scoring
- Security requirements
- OWASP AI security resources and MITRE ATLAS concepts

Hands-On Lab Objective

Threat-model a local RAG application and produce a prioritized mitigation list.

Practical Lab Activities

40. Inventory assets.
41. Identify entry points.
42. Create misuse cases.

43. Map selected threats to OWASP/MITRE resources.
44. Choose controls.
45. Review residual risk.

Free / Local Tools

OWASP resources, MITRE ATLAS, diagrams.net, LibreOffice

Expected Learning Outcome

Students can conduct a practical threat model for an AI system.

Module 09: Prompt Injection & AI Application Attacks

Theory Concepts

- Direct and indirect prompt injection
- Jailbreak concepts
- Context manipulation
- Insecure output handling
- Sensitive-information disclosure
- Model denial-of-service concepts
- Excessive trust in model output

Hands-On Lab Objective

Test an intentionally vulnerable local AI application and demonstrate safe failure modes.

Practical Lab Activities

46. Create benign injection test cases.
47. Test whether untrusted text changes instructions.
48. Observe unsafe output handling.
49. Add input separation and output validation.
50. Retest.
51. Document evidence.

Free / Local Tools

Ollama, Python, Local test harness, OWASP guidance

Expected Learning Outcome

Students can recognize AI application attack patterns and implement basic mitigations.

Module 10: AI Access Control & Guardrails

Theory Concepts

- Least privilege
- Model/data/agent permissions
- Role-based access
- Rate and token limits
- Input/output validation
- Prompt templates
- Human approval gates

Hands-On Lab Objective

Add security controls to an AI assistant before allowing it to access tools.

Practical Lab Activities

52. Create analyst/admin roles.
53. Restrict data access.
54. Add rate limits.
55. Validate tool parameters.
56. Require approval for sensitive actions.
57. Log denied actions.
58. Test privilege boundaries.

Free / Local Tools

Python, FastAPI or Flask, Ollama, SQLite

Expected Learning Outcome

Students can implement basic guardrails and least-privilege controls.

Module 11: AI Monitoring, Logging & Evaluation

Theory Concepts

- Prompt/response logging
- Privacy-aware logging
- Confidence and uncertainty
- Accuracy and hallucination evaluation
- Bias and consistency checks
- Usage/rate monitoring
- Audit trails

Hands-On Lab Objective

Create a local evaluation harness for a security assistant.

Practical Lab Activities

59. Build a test dataset.
60. Run the same questions repeatedly.
61. Score accuracy and groundedness.
62. Detect hallucinated indicators.
63. Record response time and usage.
64. Generate an evaluation report.

Free / Local Tools

Python, Pandas, JSON/CSV, Ollama

Expected Learning Outcome

Students can measure AI security quality instead of trusting output blindly.

Module 12: AI-Assisted SOC Operations

Theory Concepts

- AI-assisted alert triage
- Log summarization
- Event correlation
- Investigation hypotheses
- Threat-hunting assistance

- Human-in-the-loop response

Hands-On Lab Objective

Use AI to assist a SOC investigation using synthetic or lab-generated logs.

Practical Lab Activities

65. Feed selected Wazuh/log events to a local model.
66. Summarize an alert.
67. Extract entities and IOCs.
68. Suggest investigation queries.
69. Compare suggestions with analyst findings.
70. Document the final human decision.

Free / Local Tools

Wazuh, Ollama, Python, Wireshark

Expected Learning Outcome

Students can integrate local AI into SOC analysis without delegating final security decisions.

Module 13: AI-Assisted Vulnerability & Code Security

Theory Concepts

- Secure coding review
- SAST concepts
- Dependency risk
- CVE interpretation
- False-positive validation
- AI-generated code risks

Hands-On Lab Objective

Use AI alongside free code-security tools to review a small vulnerable application.

Practical Lab Activities

71. Run Bandit or Semgrep.
72. Review findings with the local model.
73. Ask AI to explain risky code.
74. Validate findings manually.
75. Fix selected issues.
76. Retest with the scanner.

Free / Local Tools

Semgrep Community, Bandit, Trivy, Ollama, Python

Expected Learning Outcome

Students can combine AI explanation with deterministic security tooling.

Module 14: AI-Assisted Threat Intelligence

Theory Concepts

- IOC extraction
- Indicator normalization

- Threat-context enrichment
- MITRE ATT&CK mapping
- Threat-intelligence confidence
- Report generation

Hands-On Lab Objective

Build a local pipeline that extracts IOCs from a simulated threat report and maps behaviors to ATT&CK.

Practical Lab Activities

77. Parse a text report.
78. Extract IPs/domains/hashes.
79. Normalize indicators.
80. Ask AI to suggest ATT&CK techniques.
81. Verify against ATT&CK.
82. Produce a structured intelligence report.

Free / Local Tools

Python, Ollama, MITRE ATT&CK, MISP Community Edition (optional)

Expected Learning Outcome

Students can accelerate threat-intelligence workflows while retaining verification.

Module 15: AI for Malware & Forensics

Theory Concepts

- Safe malware-analysis principles
- Static analysis
- Hashes, strings and metadata
- YARA concepts
- Forensic timelines
- AI-assisted evidence summarization

Hands-On Lab Objective

Use AI to organize evidence from benign/safe samples and a provided forensic dataset.

Practical Lab Activities

83. Calculate hashes.
84. Extract metadata and strings.
85. Run a YARA rule.
86. Summarize evidence with AI.
87. Check every AI claim against source evidence.
88. Create a timeline.

Free / Local Tools

YARA, ClamAV, Autopsy, Volatility 3, Ollama

Expected Learning Outcome

Students can use AI for evidence organization without treating generated conclusions as proof.

Module 16: Security Automation with Python

Theory Concepts

- Python security scripting
- Regex and parsing
- JSON/CSV processing
- API clients
- Error handling
- Logging
- Safe automation boundaries

Hands-On Lab Objective

Build a Python tool that reads logs, extracts IOCs and creates a structured analyst report.

Practical Lab Activities

89. Parse a local log.
90. Extract indicators.
91. Validate values.
92. Enrich from a local lookup table.
93. Call a local LLM.
94. Generate JSON/Markdown output.
95. Add logging and error handling.

Free / Local Tools

Python, Ollama, Pandas

Expected Learning Outcome

Students can build a practical AI-assisted security automation.

Module 17: AI Agents & Tool Calling

Theory Concepts

- Agent loop: goal, reasoning, action and observation
- Tool schemas
- Function calling concepts
- State and memory
- Permission boundaries
- Human approval
- Agent failure modes

Hands-On Lab Objective

Build a read-only security assistant with safe local tools.

Practical Lab Activities

96. Create check_ip-style local lookup logic using test data.
97. Create log-analysis tool.
98. Define strict tool inputs.
99. Prevent arbitrary shell execution.
100. Add approval before any sensitive action.
101. Record tool calls.

Free / Local Tools

Python, Ollama, SQLite/JSON, VS Code

Expected Learning Outcome

Students understand how agents interact with tools and why excessive agency is dangerous.

Module 18: MCP-Style Security Integrations

Theory Concepts

- Model-to-tool integration concepts
- Tool descriptions and schemas
- Read-only security tools
- Input validation
- Connector trust
- Least privilege

Hands-On Lab Objective

Build a small local security tool server with safe read-only functions.

Practical Lab Activities

102. Create check_ip using local/synthetic data.
103. Create search_cve using a local dataset.
104. Create analyze_log.
105. Validate all inputs.
106. Return structured results.
107. Connect the assistant to the tools.
108. Test malicious tool parameters.

Free / Local Tools

Python, MCP Python SDK where available, Ollama, JSON

Expected Learning Outcome

Students can create a safe foundation for AI-to-security-tool integration.

Module 19: AI-Enhanced Attack Simulation

Theory Concepts

- AI-assisted reconnaissance concepts
- Automated social-engineering risks
- Synthetic deepfake/impersonation concepts
- Obfuscation and automated attack generation
- Defensive countermeasures

Hands-On Lab Objective

Study how AI can accelerate attacks without targeting real people or systems.

Practical Lab Activities

109. Generate a synthetic phishing example.
110. Identify AI-assisted indicators.
111. Build detection rules/checklists.
112. Compare human vs AI-generated reconnaissance notes.

113. Create defensive awareness controls.

Free / Local Tools

Ollama, Python, Wazuh, Sigma concepts

Expected Learning Outcome

Students understand the changing attack surface while keeping exercises controlled and defensive.

Module 20: AI Governance, Risk & Responsible Use

Theory Concepts

- AI policy and acceptable-use rules
- Public vs private models
- Data classification
- Privacy and intellectual-property considerations
- Bias and fairness
- Transparency and accountability
- AI risk management
- Human oversight

Hands-On Lab Objective

Create an AI security policy for a fictional organization.

Practical Lab Activities

114. Define permitted AI uses.
115. Classify sensitive data.
116. Define prohibited inputs.
117. Create approval rules.
118. Define logging/retention.
119. Create an AI incident escalation path.
120. Map controls to NIST AI RMF concepts.

Free / Local Tools

LibreOffice, NIST AI RMF reference material

Expected Learning Outcome

Students can connect technical AI security controls with organizational governance.

Module 21: AI Security Operations Challenge

Theory Concepts

- Alert triage
- Prompt abuse investigation
- Data leakage analysis
- Agent/tool audit
- Threat intelligence
- Containment and lessons learned

Hands-On Lab Objective

Investigate a simulated incident involving an AI assistant, compromised data and suspicious tool usage.

Practical Lab Activities

121. Review prompts and logs.
122. Identify the initial failure.
123. Trace data flow.
124. Review tool calls.
125. Determine affected assets.
126. Recommend containment.
127. Harden the system.
128. Produce an incident report.

Free / Local Tools

Wazuh, Python, Ollama, Wireshark, LibreOffice

Expected Learning Outcome

Students demonstrate end-to-end AI security investigation and response.

Module 22: Final Capstone: Secure AI Cybersecurity Assistant

Theory Concepts

- Requirements and architecture
- RAG
- Security controls
- AI evaluation
- Tool integration
- Monitoring
- Governance
- Demonstration

Hands-On Lab Objective

Build a portfolio-ready local AI cybersecurity assistant using free tools.

Practical Lab Activities

129. Define the security use case.
130. Build a local LLM interface.
131. Add a small RAG knowledge base.
132. Implement at least two read-only security tools.
133. Add input/output validation.
134. Add logging.
135. Create an evaluation dataset.
136. Perform AI security tests.
137. Document threats and controls.
138. Demonstrate the assistant and explain its limitations.

Free / Local Tools

Ollama, Python, ChromaDB/FAISS, Wazuh, MITRE ATT&CK, OWASP resources, MCP Python SDK (optional)

Expected Learning Outcome

Students finish with a working, documented AI-security project that demonstrates both AI usage and AI protection skills.

Assessment Structure

Component	Weight	Evidence
AI & Security Knowledge Checks	15%	Short quizzes and concept checks
Hands-On Labs	25%	Completed lab evidence and configurations
AI Security Testing Exercise	15%	Test an AI application and implement controls
SOC Automation Challenge	15%	Build an AI-assisted security workflow
Final Capstone	30%	Build, secure, evaluate and demonstrate the local AI cybersecurity assistant

Final Capstone Deliverables

- Architecture/data-flow diagram
- Threat model and risk register
- Local model setup documentation
- RAG knowledge base
- At least two safe read-only security tools
- Prompt-injection/security test results
- Access-control and guardrail implementation
- Evaluation dataset and quality report
- Monitoring/audit log
- User guide and limitations
- Final demonstration

Course Safety Rules

- Only use systems, applications, accounts and datasets that the learner owns or has explicit authorization to test.
- Do not submit confidential company data, credentials, private keys or personal information to an AI model.
- Use synthetic logs, test accounts and intentionally vulnerable applications for attack simulations.
- Keep vulnerable applications and security-testing VMs isolated.
- Do not give AI agents unrestricted shell, browser, filesystem or network privileges.
- Prefer read-only tools during beginner agent labs.
- Every AI-generated security conclusion must be treated as a suggestion and verified against source evidence.
- Do not use AI-generated payloads or attack automation against public systems.

Suggested Course Positioning

Suggested tagline: “LEARN AI → SECURE AI → AUTOMATE SECURITY → BUILD AI SECURITY AGENTS”

This course can be positioned as an independent practical AI-cybersecurity program. It is especially suitable for SOC analysts, security engineers, ethical hackers, developers moving into security, cybersecurity students and security leaders who want a hands-on introduction to AI-enabled defense.

Reference & Current-Status Note

CompTIA's publicly available SecAI+ CY0-001 document reviewed for this curriculum is explicitly labeled “Pre-draft Exam Objectives.” CompTIA states that pre-draft objectives provide directional information and may change during development. Therefore this document does not claim to reproduce or guarantee coverage of a final certification exam.

The official reference reviewed includes broad areas covering AI concepts, securing AI systems, AI-assisted security, and AI governance/risk/compliance. This course converts those broad themes into an original, free-tool practical curriculum.